

- ▶ Assist in troubleshooting and resolving tools / pipeline related issues.
- ▶ Support game teams in feature development.

Must-have skills for a Tools Programmer

- ▶ Excellent C# understanding and skills. The ability to write highly efficient and optimized code.
- ▶ Strong understanding of the game engine.
- ▶ Knowledge of the target systems (PC, or one of the consoles).
- ▶ Solid understanding of C/C++, possibly Assembly, mathematical concepts, graphics, collision detection, object oriented programming, and database management.
- ▶ Good communication skills so as to understand the requirement of the artists and designers
- ▶ Knowledge of user interface design as a tool is great if it has a handy interface.
- ▶ Clear and neat documentation skills.
- ▶ Excellent debugging skills.

Graphic Programmer

A Graphics Programmer in particular must master the full suite of techniques to realise three dimensional objects inside a two dimensional display and must develop programs to handle complex 2D and 3D graphics.

What does a Graphic Programmer do?

- ▶ Develop core graphic engine and rendering systems
- ▶ Designing and implementing systems and tools to support rendering and technology needs (such as lighting, occlusion, shaders).
- ▶ Implement realistic and fascinating particle effects.
- ▶ Implement/improve visual effects using current in house technology.
- ▶ Collaborate with Art Direction and Design to find the best ways of generating great graphics with minimal performance costs.

Must-have skills for a Graphic Programmer

- ▶ Math expertise (especially linear algebra and advanced calculus).
- ▶ Expert in modern rendering and engine-related issues.
- ▶ Knowledge of writing shaders.
- ▶ Excellent code/performance optimization skills.
- ▶ Knowledge of graphic APIs